

Embrace Edge AI, Empower the era of visual recognition



EC4600 Series AI Edge Computer

· 14 TOPS AI

· 4× Camera

· Rich Interfaces

· High Reliability

1. Product Overview

EC4600 AI edge computer for real-time visual recognition and local image analytics.

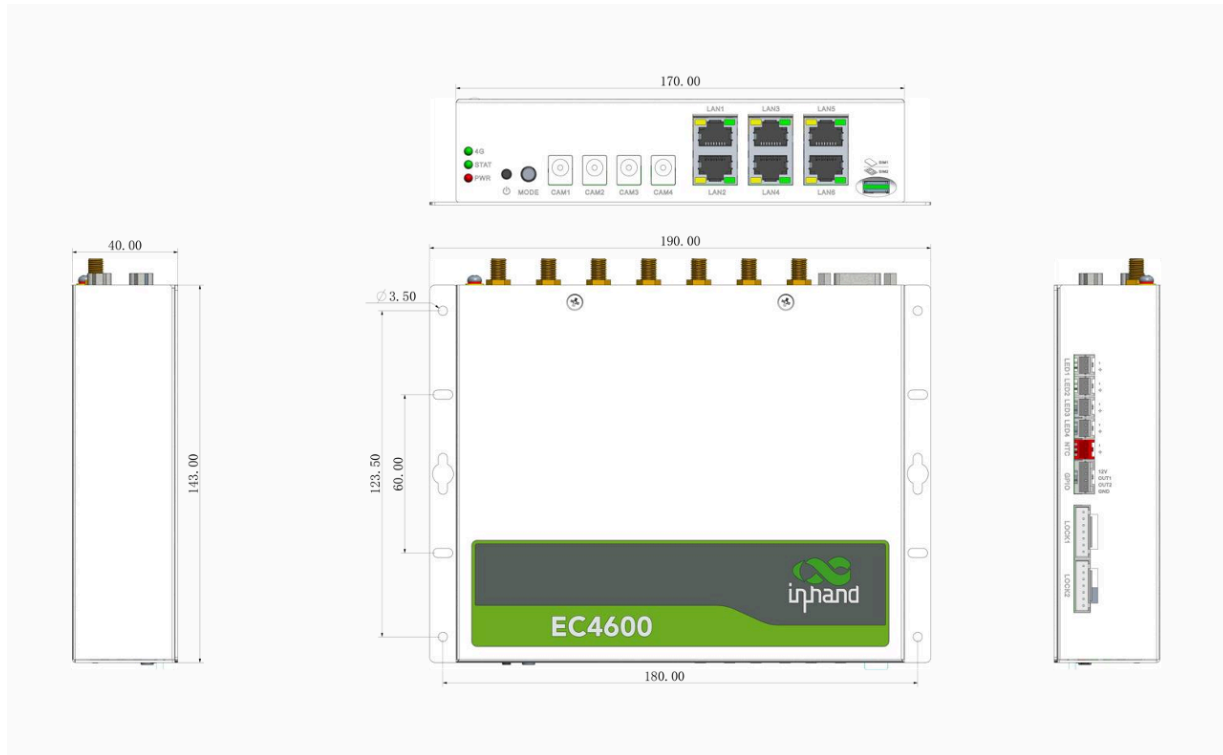
Key features:

- **Edge AI compute:** 14 TOPS, octa-core ARM, Adreno GPU
- **Vision-ready I/O:** 4× camera power, HDMI, USB, SPK
- **Flexible connectivity:** 4G, Wi-Fi 6, 6× FE, dual SIM
- **Industrial design:** Metal shell, IP40, fanless, wide temp
- **Stable operation:** HW/SW watchdog, link detect, BAT

Core Technical Specifications

Specification Item	Value
Cellular Network	LTE Cat4, 4G; 2 × Nano SIM; dual SIM failover
Network Features	ARP, Ethernet; Static IP, DHCP
Security	Secure Boot
Operating System	Android 12
Reliability	Dual SIM backup; multi-level link detection; embedded watchdog
Wi-Fi	STA; 802.11a/b/g/n/ac/ax; 2.4 / 5 GHz dual band
AI Computing Power	14 TOPS (comprehensive); octa-core ARM up to 2.7 GHz
Dimensions (W × D × H)	190 × 143 × 40 mm (including mounting parts); wall mounting
Ethernet Ports	6 × 10/100 Mbps
Vision & Serial I/O	4 × camera power (5 W); 2 × RS-232; 2 × RS-485; HDMI; 2 × USB 2.0
Power Supply	12 V DC
Operating Temperature & Protection	-10 °C ~ +60 °C; IP40; fanless metal housing

2. Product Dimensions



Top View

Note:

1. All dimensions are in millimeters (mm).
2. Dimensions (W × D × H): 190 × 143 × 40 mm (including installation parts).
3. All dimensions are approximate, **for reference only**.
4. Dimensions shown **shall not be used for production**.
5. Dimensions are subject to part and manufacturing tolerances.
6. Specifications may change without prior notice.

3. Hardware Specifications

Category/Parameter	Specification
Hardware Performance	
CPU	Qualcomm QCM6490: 1 × Kryo 670 Gold Plus @ 2.7 GHz + 3 × Kryo 670 Gold @ 2.4 GHz + 4 × Kryo 670 Silver @ 1.9 GHz
GPU	Qualcomm® Adreno™ 643 @ 812 MHz
AI Computing Power	14 TOPS (comprehensive)
RAM	8 GB
Flash	128 GB eMMC

Category/Parameter	Specification
Hardware Interface	
Ethernet Port	6 × 10/100 Mbps
Camera Power Supply	4 × white round interface (Ø5.5 × 2.1), 5 W
Serial Port	2 × RS-232 (DB9); 2 × RS-485 (one 4-pin with power supply, one 3-pin industrial terminal)
GPIO	2 × OUT
NTC	1 × NTC temperature sensing
USB	2 × USB 2.0
HDMI	1 × HDMI
Speaker	1 × SPK, supports 5 W speakers
Backup Battery	1 × backup battery interface
Power Button	1 × Power
SIM	2 × Nano SIM slot
Wi-Fi	
Operating Mode	STA
Protocol	802.11a/b/g/n/ac/ax
Frequency Band	2.4 / 5 GHz dual band
LED Indicators	
Power	Power supply indicator, always on after power-on
Status	Status indicator flashes during normal operation
3G/4G	Cellular indicator always on when connected normally
User LED	4 × LED
Power	
Power Input	12 V DC
Mechanical	
Mounting	Wall mounting
Ingress Protection	IP40
Dimensions (W × D × H)	190 × 143 × 40 mm (including installation parts)

Category/Parameter	Specification
Housing	Metal
Cooling	Fanless
Environmental	
Operating Temperature	-10 °C ~ +60 °C
Storage Temperature	-40 °C ~ +85 °C
Ambient Humidity	5 ~ 95 % RH (non-condensing)
Physical	
ESD	Level 2
EFT	Level 2
Surge	Level 2
Shock	IEC 60068-2-27
Vibration	IEC 60068-2-6
Free Fall	IEC 60068-2-32

4. Software Specifications

Category/Parameter	Specification
Operating System	
OS	Android 12
Network Features	
Network Type	4G
LAN Protocol	ARP, Ethernet
WAN Protocol	Static IP, DHCP
Security	
Secure Boot	Supported
Reliability	
Backup	Dual SIM
Link Detection	Multi-level link detection; auto redials once disconnected
Embedded Watchdog	Device self-diagnosing; auto-recovers from operation faults

5. Ordering Information

Model Rule

Model code: EC4600-<WMNN>-<STD>

<WMNN>: Cellular Type & Frequency Band

Optional OS suffix per datasheet: -[X] (OS optional)

Model List

Model	Region	<WMNN>: Cellular Type & Frequency Band	RAM	Flash	Ethernet Port	Serial Port
EC4600-DQ20-STD	China	LTE Cat4 LTE-FDD: B1/B3/B5/B8 LTE-TDD: B34/B38/B39/B40/B41 WCDMA: B1/B8 TD-SCDMA: B34/B39 CDMA: BC0 GSM: 900/1800 MHz	8 GB	128 GB	6 × 100 Mbps	2 × RS-232 2 × RS-485
EC4600-FQ58-STD	EMEA	LTE Cat4 LTE-FDD: B1/B3/B7/B8/B20/B28A LTE-TDD: B38/B40/B41 WCDMA: B1/B8 GSM: B3/B8	8 GB	128 GB	6 × 100 Mbps	2 × RS-232 2 × RS-485

6. Contact Us

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